

LAL BAHADUR SHASTRI INTL, India

WMO: 424790

Lat: 25.450N

Lon: 82.867E

Elev: 81

StdP: 100.35

Time Zone: 5.50 (IND)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	7.2	8.8	3.2	4.8	26.6	4.9	5.4	22.0	5.4	20.2	4.9	20.4	0.7	270	0.357

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.3	43.0	23.1	41.9	23.4	40.2	23.7	29.0	34.0	28.7	33.4	28.3	32.7	4.1	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
28.0	24.4	31.1	27.5	23.6	30.7	27.2	23.1	30.5	95.7	34.0	93.9	33.5	92.2	33.0	34.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.8	5.7	5.0	4.9	44.9	1.7	0.8	3.7	45.5	2.7	46.0	1.8	46.4	0.6	47.0		
			4.6	30.1	1.4	1.1	3.5	30.9	2.7	31.6	1.9	32.2	0.9	33.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	26.3	15.5	19.6	25.3	31.2	33.5	33.3	30.3	30.0	29.5	27.4	22.5	17.2	(6)
(7)		DBStd	6.41	2.62	2.51	2.95	2.40	2.08	2.81	1.91	1.35	1.52	2.07	2.15	2.58	(7)
(8)		HDD10.0	2	1	0	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	158	93	13	0	0	0	0	0	0	0	0	1	51	(9)
(10)		CDD10.0	5948	170	269	473	635	729	700	631	621	586	538	374	222	(10)
(11)		CDD18.3	3064	4	49	215	385	471	450	372	362	336	279	124	15	(11)
(12)		CDH23.3	40766	48	468	2453	5640	7428	6974	4916	4676	4159	2956	926	121	(12)
(13)	CDH26.7	22954	4	113	1299	3656	5016	4609	2495	2246	1910	1323	276	7	(13)	
(14)	Wind	WSAvg	1.7	1.3	1.5	1.8	2.1	2.5	2.5	2.1	1.9	1.8	1.1	0.9	1.0	(14)
(15)	Precipitation	PrecAvg	1133	17	20	13	14	45	160	310	300	203	41	4	5	(15)
(16)		PrecMax	1449	57	73	59	51	102	390	498	436	365	160	50	56	(16)
(17)		PrecMin	822	0	1	0	0	3	51	131	183	57	1	0	0	(17)
(18)		PrecStd	165	14	21	12	11	26	86	86	71	82	39	10	12	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.9	32.1	39.2	43.8	44.2	44.2	38.9	36.2	36.1	35.2	32.1	27.8	(19)
(20)		0.4%	MCWB	17.4	19.7	20.9	20.4	22.9	24.3	27.5	28.2	27.8	25.7	22.1	18.0	(20)
(21)		2%	DB	24.8	29.8	36.9	42.1	43.0	43.0	36.9	35.1	35.1	34.2	30.9	26.1	(21)
(22)		2%	MCWB	16.7	19.7	20.9	20.9	23.3	24.6	27.7	28.2	27.6	25.4	21.1	17.3	(22)
(23)		5%	DB	23.0	28.0	35.0	40.9	41.9	41.2	35.2	34.1	34.1	33.2	29.9	24.9	(23)
(24)		5%	MCWB	15.6	18.5	20.5	21.2	23.6	25.7	27.7	28.1	27.5	24.9	20.7	16.8	(24)
(25)		10%	DB	21.6	26.2	33.1	39.2	40.2	39.7	34.0	33.2	33.2	32.2	28.2	23.2	(25)
(26)		10%	MCWB	15.3	17.6	20.1	21.3	24.1	26.0	27.9	27.8	27.3	24.3	20.0	16.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.2	21.7	23.7	26.2	28.4	29.2	29.4	29.6	29.2	28.1	24.2	20.1	(27)
(28)		0.4%	MCDWB	24.0	27.9	33.7	36.3	38.6	36.9	34.0	33.7	34.1	32.3	29.0	25.1	(28)
(29)		2%	WB	18.0	20.6	22.3	24.9	27.5	28.6	28.9	29.0	28.6	27.2	22.9	18.9	(29)
(30)		2%	MCDWB	22.1	27.4	32.7	36.0	37.2	35.9	33.4	33.3	33.1	31.6	28.3	23.9	(30)
(31)		5%	WB	17.0	19.7	21.4	24.1	26.8	28.0	28.5	28.6	28.2	26.6	22.1	17.9	(31)
(32)		5%	MCDWB	20.5	26.3	31.6	35.0	36.1	34.9	32.8	32.5	32.1	30.8	27.1	22.5	(32)
(33)		10%	WB	16.1	18.7	20.7	23.1	26.1	27.5	28.2	28.2	27.7	26.0	21.1	17.0	(33)
(34)		10%	MCDWB	20.3	24.5	30.6	33.7	35.2	34.3	32.1	31.8	31.4	30.1	26.5	21.4	(34)
(35)	Mean Daily Temperature Range	MDBR	11.3	12.9	15.0	15.9	13.3	10.2	6.7	6.5	7.1	10.3	12.5	11.8	(35)	
(36)		5% DB	MCDBR	13.8	14.6	16.8	17.4	15.3	13.2	8.6	7.7	8.7	10.9	12.9	13.7	(36)
(37)		5% DB	MCWBR	6.6	6.1	5.0	4.7	4.5	3.6	2.7	2.7	2.9	3.5	4.6	5.9	(37)
(38)		5% WB	MCDBR	11.1	13.1	15.0	14.4	12.4	9.6	7.2	7.0	7.5	8.7	10.9	11.5	(38)
(39)		5% WB	MCWBR	6.2	6.1	5.2	5.0	4.4	3.3	2.9	2.9	3.0	3.3	4.6	5.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.571	0.530	0.526	0.598	0.694	0.730	0.646	0.605	0.590	0.629	0.677	0.650	(40)	
(41)		taud	1.704	1.850	1.888	1.767	1.615	1.585	1.767	1.889	1.898	1.742	1.576	1.592	(41)	
(42)		Ebn at Noon	663	746	779	735	665	638	692	721	718	655	571	572	(42)	
(43)		Edh at Noon	211	195	197	228	265	271	226	199	193	215	239	229	(43)	
(44)	All-Sky Solar Radiation	RadAvg	3.12	4.46	5.81	6.52	6.46	5.49	4.46	4.64	4.64	4.55	3.65	2.98	(44)	
(45)		RadStd	0.43	0.33	0.24	0.27	0.28	0.48	0.41	0.24	0.37	0.32	0.20	0.30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page